Gotu AI Agriculture

Where Tech with Intelligence Meets Soil

1 Main Topic: [Banana] – [Harvesting]

1.1 Subtopic 1: Maturity Indices and Pre-Harvest Field Guidelines

1.1.1 1. Visual and Temporal Maturity Indices

Harvest readiness in commercial export banana (*Musa* spp.) production is determined through precise physical and temporal parameters [Banana Expert System, Do's and Don'ts #15; Banana$_d$*isease*$_d$*etection.*

- **Finger Roundness and Angularity:** Fruits are harvested at the $3/4th$ maturity stage (full green stage) when prominent longitudinal ridges and sharp angularities become smooth and rounded [Banana Expert System, Do's and Don'ts #15].
- **Color Shift:** The peel color transitions from dark green to a uniform light-green shade [Banana Expert System, Do's and Don'ts #15].
- **Temporal Duration:** Typically, 80 to 100 days elapse between inflorescence shooting/bunch emergence and harvest maturity, depending on ambient temperature, seasonal solar radiation, and cultivar [Banana Expert System, Tagging].
- **Blemish Rejection Standards:** Bunches selected for export must be whole, green, and free from leaf scratching, rubbing, sunburn, or insect scarring [Banana Expert System, Do's and D
  Reject bunches displaying malformed fingers, octopus-shaped hands, or split/torn peel [Banana Expert System, Do's and Don'ts #16].

1.1.2 2. Pre-Harvest Hydrological Cut-Off

- **Irrigation Cut-Off Schedule:** Irrigation must be stopped strictly 7 to 10 days (preferably 1 week) prior to the planned harvest date [Banana Expert System, Do's and Don'ts #13].
- **Operational and Pathological Purpose:** Terminating pre-harvest irrigation allows soil surface drying to facilitate field labor movement, prevent soil compaction from haulage trailers, and minimize fruit water-soaking, latex staining, or peel splitting during post-harvest transport [Banana Expert System, Do's and Don'ts #13].

1.1.3 3. Pre-Harvest Pesticide Clearance and Pre-Harvest Interval (PHI)

- **Chemical Spray Cut-Off:** All chemical pesticide applications must cease at least 15 days prior to bunch harvesting [IPM-Banana-Revised-Sept2011.pdf, p. 54, 57].
- **Export Residue Compliance:** Strictly observing the Pre-Harvest Interval (PHI) eliminates chemical pesticide residues in the fruit pulp, ensuring full compliance with international Maximum Residue Limits (MRLs) and sanitary export standards [Banana.pdf, p. 42; IPM-Banana-Revised-Sept2011.pdf, p. 57].

1.1.4 4. Functional Canopy Leaf Threshold at Harvest

- **Active Green Leaf Count:** A minimum of 7 to 10 active, functional green leaves must remain intact on the pseudostem at the exact time of harvest [Banana Expert System, Removal of withere
- **Green Life and Shelf-Life Impact:** Maintaining adequate functional green foliage up to harvest ensures optimal carbohydrate reserves, prevents premature ripening in transit, and extends the post-harvest green life of exported bunches during cold storage [Banana Expert System, Removal of withered styles].

1.2 Subtopic 2: Harvesting Operations and Field Safety/Handling

1.2.1 1. Harvesting Protocol and Team Operations

- **Two-Person Harvesting Team:** Harvesting must be carried out by a trained two-person team consisting of a cutter and a bearer [Banana Expert System, Harvesting; Banana$_d$*isease*$_d$*etection.pdf*, *Sec*
- **Controlled Pseudostem Lowering:** The cutter makes a partial transverse slant cut into the pseudostem 1.5 to 2.0 meters above ground level on the side toward which the bunch hangs [Banana Expert System, Harvesting]. The weight of the heavy bunch causes the top of the pseudostem to bend slowly downward without snapping, allowing the bearer to support and catch the bunch on a padded shoulder cushion, stretcher, or foam cradle before it touches the soil [Banana Expert System, Harvesting; Banana$_d$*isease*$_d$*etection.pdf*,
- **Impact Injury Prevention:** Bunches must **never** be dropped directly onto the soil or allowed to strike hard surfaces, as impact bruises cause immediate internal tissue softening, latex staining, and rapid rot development [Banana Expert System, Do's and Don'ts #16; Banana$_d$*is*

1.2.2 2. Peduncle Severing Specifications

- **Clean Peduncle Cut:** Sever the main peduncle in a single clean stroke using a sharp, disinfected harvesting knife [Banana Expert System, Harvesting].
- **Severing Distance:** Cut the peduncle **20 cm to 25 cm above the first hand** (leaving a 7.5 cm to 10 cm stalk clear of fingers) [Banana Expert System, Harvesting; Banana$_d$*isease*$_d$*etection.pdf*, *Se*
  Retaining an adequate peduncle stalk handle facilitates manual handling, hanging on overhead cableways, and post-harvest de-handing without damaging the top fruit hand [Banana Expert System, Harvesting].

1.2.3 3. Latex Bleeding and Field Transport Standards

- **Latex Bleeding Phase:** Immediately after severing, stand harvested bunches upright on a clean, padded surface or dry leaf trash layer for 15 minutes to allow raw latex exudation to stop [Banana Expert System, Harvesting; Banana$_d$*isease*$_d$*etection.pdf*, *Section*16, *p.*25].
  Preventing sticky latex from dripping onto fruit fingers eliminates permanent dark latex staining and skin blemishes [Banana Expert System, Harvesting].
- **Padded Field Transit:** Transport harvested bunches from field to pack-house using padded stretchers, foam-lined field trailers, or overhead cableway systems [Banana Expert System, Harvesting; Banana$_d$*isease*$_d$*etection.pdf*, *Section*16, *p.*25]. Direct contact between fruit fingers and soil or rough trailer walls must be strictly prevented to eliminate soil-borne fungal contamination and skin abrasions [Banana Expert System, Do's and Don'ts #16; Banana$_d$*isease*$_d$

1.3 Subtopic 3: Pack House Post-Harvest Processing (De-handing, Washing, & Disease Control)

1.3.1 1. De-handing and Sorting

- **De-handing Technique:** Sever hands cleanly from the main peduncle using a sharp, sanitized curved de-handing knife under shaded pack-house facilities [Banana$_d$*isease$_d$etection.pdf*, *S*
- **Crown Retention:** Retain a smooth 2 cm to 3 cm crown base on each hand [Banana$_d$*isease$_d$etection.pdf*, *Section*16, *p.*25]. Retaining an intact crown base prevents individual finger detachment ('finger drop') and provides structural support during transit packaging [Banana$_d$*isease*$_1$*.pdf*, *p.*3; *Banana$_d$isease$_d$etection.pdf*, *Section*16, *p.*25].
- **Initial Sorting:** Immediately discard immature, overripe, split, or insect-scarred fingers during de-handing [Banana Expert System, Do's and Don'ts #16; Banana$_d$*isease$_d$etection.pdf*, *Section*16, *p.*26].

1.3.2 2. Curing, Latex Washing, and Thermal Field Heat Reduction

- **Curing and Field Heat Removal:** Spread harvested fruit on padded straw cushions or field yard tables to allow curing and bring down field heat ($30\,^{\circ}C$ to $35\,^{\circ}C$), stabilizing post-harvest respiration rates [Banana$_d$*isease$_d$etection.pdf*, *Section*16, *p.*25, 26].
- **Latex Washing Tanks:** Submerge de-handed clusters into continuous-flow washing tanks containing dilute Sodium Hypochlorite (10 to 100 ppm) or Alum (aluminum sulphate) [Banana$_d$*isease$_d$etection.pdf*, *Section*16, *p.*25; *IPM − Banana − Revised − Sept*2011*.pdf*, *p.*34].
- **Surface Cleaning and Lustre Restoration:** Washing strips raw latex exudate from cut crown tissues, removes field dirt, and restores the natural light-green skin lustre required for premium export presentation [Banana$_d$*isease$_d$etection.pdf*, *Section*16, *p.*25].

1.3.3 3. Post-Harvest Fungicidal Drenching and Disease Management

To prevent post-harvest crown decay during sea shipment, hands are treated with targeted fungicides [Banana$_d$*isease*$_1$*.pdf*, *p.*3; *IPM − Banana − Revised − Sept*2011*.pdf*, *p.*55, 57]:

- **Fungicidal Drench / Dip Application:** Dip or drench washed hands in a fungicide bath containing Thiabendazole, Benomyl, Carbendazim (0.1% / 1.0 g/L), or Prochloraz (0.1%) [Banana$_d$*isease*$_1$*.pdf*, *p.*3; *IPM − Banana − Revised − Sept*2011*.pdf*, *p.*55]. Alternatively, insert fungicide-impregnated cellulose pads directly over crown cuts during carton packing [Banana$_d$*isease*$_1$*.pdf*, *p.*3].
- **Targeted Pathogen Spectrum:** Fungicidal drenching suppresses:
  - *Crown Rot Complex: Colletotrichum musae*, *Fusarium* spp., *Verticillium theobromae*, and *Botryodiplodia theobromae* [Banana$_d$*isease*$_1$*.pdf*, *p.*3].
  - *Anthracnose: Colletotrichum musae* [Banana$_d$*isease*$_1$*.pdf*, *p.*3; *IPM − Banana − Revised − Sept*2011*.pdf*, *p.*55].
  - *Stem-End Rot: Thielaviopsis paradoxa* [Banana$_d$*isease*$_1$*.pdf*, *p.*3].
- **Alternative Non-Chemical Treatments:** Hot water immersion ($50\,^{\circ}C$ for 5 minutes) or dipping in 50% water extract of *Solanum torvum* effectively controls anthracnose and stem-end rot in organic export production [Banana$_d$*isease*$_1$*.pdf*, *p.*3; *IPM − Banana − Revised −*

1.3.4 4. Air-Drying and Protective Surface Waxing

- **Forced Air-Drying:** Transfer washed, treated hands onto padded drying racks equipped with forced-air blowers to completely evaporate free surface moisture from crown cuts and finger skins [Banana$_d$isease$_d$etection.pdf, Section16, p.25]. Packing wet fruit induces fungal mold proliferation inside cartons [Banana$_d$isease$_1$.pdf, p.3; Banana$_d$isease$_d$etection.pdf, Section16, p.25].
- **Surface Waxing:** Apply a food-grade wax emulsion integrated with Benomyl over dried hands [IPM-Banana-Revised-Sept2011.pdf, p. 55]. Wax coating seals cut vascular vessels, retards moisture transpiration, delays ethylene-mediated ripening, and extends total post-harvest green life during cold transport [IPM-Banana-Revised-Sept2011.pdf, p. 55].

1.4 Subtopic 4: Grading, Packaging, and Export Quality Specifications

1.4.1 1. Size, Caliber, and Blemish Quality Grading

- **Export Grading Standards:** Hands and clusters are sorted into standardized caliber categories based on finger length and girth (diameter) [Banana$_d$isease$_d$etection.pdf, Section16, p.26]:
  - *Extra-Large Grade:* Finger length exceeding 20 cm with a minimum diameter/caliber of 39 mm to 46 mm [Banana$_d$isease$_d$etection.pdf, Section16, p.26].
  - *Large Grade:* Finger length between 18 cm and 20 cm with a caliber of 34 mm to 38 mm [Banana$_d$isease$_d$etection.pdf, Section16, p.26].
  - *Medium Grade:* Finger length between 15 cm and 18 cm [Banana$_d$isease$_d$etection.pdf, Section16, p.26].
- **Visual Blemish Categorization:** Sub-divide graded clusters into Premium (Grade No. 1) and Choice (Grade No. 2) quality tiers [Banana Expert System, Do's and Don'ts #15; Banana$_d$isease$_d$etectio
Grade No. 1 demands defect-free, smooth, light-green peel with zero leaf scratching, insect scarring, or latex staining [Banana Expert System, Do's and Don'ts #15].

1.4.2 2. Strict Rejection and Defect Thresholds

Hands or clusters failing export standards must be systematically discarded from export consignment streams [Banana Expert System, Do's and Don'ts #16; Banana$_d$isease$_d$etection.pdf, Section16, p.26]:

- **Morphological Deformities:** Reject hands exhibiting octopus-shaped or fused fingers, double fingers, or incomplete finger filling [Banana Expert System, Do's and Don'ts #16].
- **Physical and Pathological Defects:** Reject clusters showing mechanical blemish scars, skin splitting, sun-scorch browning, viral chlorotic streaking, or post-harvest fungal mold at the crown [Banana Expert System, Do's and Don'ts #16; Banana$_d$isease$_d$etection.pdf, Section16, p.26].
- **Maturity Deviations:** Reject overripe (yellowing) or severely thin, under-developed fingers [Banana Expert System, Do's and Don'ts #15, #16].

1.4.3 3. Export Carton Packaging and Cushioning Specifications

Export banana packaging must withstand long-distance marine transport vibrations and stacking pressures [Banana$_d$isease$_d$etection.pdf, Section16, p.26; IPM − Banana − Revised − Sept2011.pdf, p.55].

- **Corrugated Fiber Board (CFB) Cartons:** Pack hands or clusters into sturdy, telescopic 5-ply Corrugated Fiber Board (CFB) cartons with a net weight capacity of **13.0 kg to 14.5 kg** [Banana$_d$isease$_d$etection.pdf, Section16, p.26; IPM − Banana − Revised − Sept2011.pdf, p.55].

- **Polyethylene Bag Liners and Vacuum Packing:** Line cartons with 0.05 mm (50-micron) thick polyethylene film bags [Banana$_d$isease$_d$etection.pdf, Section16, p.26; IPM − Banana − Revised − Se
For long-distance sea transport, employ modified atmosphere packaging (MAP) or vacuum-sealing after inserting an ethylene absorbent pad (potassium permanganate) into the liner [Banana$_d$isease$_d$etection.pdf, Section16, p.26; IPM − Banana − Revised − Sept2011.pdf, p.55].

- **Inter-Layer Cushioning Protection:** Place soft foam sheets or bubble-wrap membranes between cluster layers inside the box, orienting finger tips inward toward the center to eliminate skin-to-skin friction, mechanical bruising, and crown pressure during container handling [Banana$_d$isease$_d$etection.pdf, Section16, p.26].

1.5 Subtopic 5: Cold Chain, Controlled Atmosphere (CA) Storage, and Ripening Protocols

1.5.1 1. Forced-Air Pre-Cooling and Rapid Heat Removal

- **Pre-Cooling Timeline:** Packed cartons must undergo forced-air pre-cooling within 10 to 12 hours of harvest [Banana$_d$isease$_d$etection.pdf, Section16, p.26].

- **Pulp Temperature Reduction:** Pre-cooling rapidly lowers fruit pulp temperature from ambient field heat ($30\,^{\circ}C$ to $35\,^{\circ}C$) down to the storage temperature of $13\,^{\circ}C$ [Banana$_d$isease$_d$etection.pdf, Section16, p.26]. Removing field heat suppresses basal respiration rates, slows moisture transpiration, and inhibits endogenous ethylene production [Banana$_d$isease$_d$etection.pdf, Section16, p.26].

1.5.2 2. Cold Storage Regimes and Chilling Injury Prevention

- **Optimum Storage Parameters:** Maintain cold storage rooms strictly at $13\,^{\circ}C$ to $14\,^{\circ}C$ with 85% to 95% relative humidity [Banana$_d$isease$_d$etection.pdf, Section16, p.26; IPM − Banana − Revised − Se

- **Chilling Injury Pathology:** Exposing green bananas to temperatures below $12\,^{\circ}C$ to $13\,^{\circ}C$ induces chilling injury [Banana$_d$isease$_d$etection.pdf, Section16, p.26]. Symptoms include sub-epidermal vascular browning, dull smoky-grey peel discoloration, surface pitting, uneven pulp softening, and complete failure to ripen normally upon exposure to ethylene [Banana$_d$isease$_d$etection.pdf, Section16, p.26].

1.5.3 3. Controlled Atmosphere (CA) Storage and Ethylene Management

- **Gas Composition Standards:** For long-distance sea transport (lasting 4 to 6 weeks), maintain a Controlled Atmosphere (CA) environment containing 2% to 5% O_2 and 2% to 5% CO_2 at $13\,^{\circ}C$ to $14\,^{\circ}C$ [Banana$_d$isease$_d$etection.pdf, Section16, p.26].

- **Ethylene Scavenging:** Keep ambient ethylene gas concentrations strictly below 1.0 ppm inside reefer containers by deploying Potassium Permanganate ($KMnO_4$) ethylene absorbent pads or catalytic scrubbers [Banana$_d$isease$_d$etection.pdf, Section16, p.26; IPM − Banana − Revis
Low oxygen combined with ethylene removal extends post-harvest green life up to 6 to 8 weeks without quality loss [Banana$_d$isease$_d$etection.pdf, Section16, p.26; IPM − Banana − Revised − Sept2011.pdf, p.

1.5.4 4. Commercial Controlled Ripening Protocols

- **Ethylene Gas Application:** Initiate uniform ripening in sealed commercial ripening chambers by trickling 100 to 1000 ppm ethylene gas for 24 to 48 hours at 15 $^{\circ}C$ to 18 $^{\circ}C$ under 85% to 90% relative humidity [Banana$_d$*isease*$_d$*etection.pdf*, *Section*16, *p.*26].
- **Calcium Carbide Prohibition:** The use of industrial Calcium Carbide (CaC_2) acetylene gas for artificial ripening is strictly prohibited due to hazardous chemical impurities (arsenic and phosphorus hydride), health risks, and inconsistent peel color development [Banana$_d$*isease*$_d$*etection.pdf*, *Section*16, *p.*26].